Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	**Zero-Lower Bound and Forward Guidance in a Behavioural New Keynesian Model with Imperfect Credibility (Structured Research Assignment)**

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Zero-Lower Bound and Forward Guidance in a Behavioural New Keynesian Model with Imperfect Credibility

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EC424 - Monetary Economics and Aggregate Fluctuations

Abstract

This essay analyses a New Keynesian Model modified with a Bayesian household that combines partially backward looking expectations with the Central bank's forecast. I show that with an imperfectly credible central bank, inflation's response to monetary policy and demand shocks is more persistent and amplified compared to that of the output gap relative to the rational expectations case. Through model simulations, I analyse how these differences relate to the way expectations of inflation and output gap evolve. I find that the welfare costs to the economy at the zero lower bound may be higher than in the rational case, and are directly proportional to the weight on past forecast errors in the household's backward looking expectation component. Finally, I show how forward guidance in this behavioural framework is weaker, suggesting a potential solution to the forward guidance puzzle through imperfect central bank credibility.

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1 Introduction

Starting with the 1990s, central banks around the world have opened up to communicating their policy decisions through statements and announcements after committee meetings. Never has this communication been more important than after the 2008 Financial Crisis. Theoretically, it is understood that most central banks cannot reduce their policy rates below a certain threshold, called the zero-lower bound (ZLB), otherwise, agents in the economy would be better off holding cash rather than saving in interest-earning assets. When interest rate based policies fail, central banks are forced to rely either on announcements of future paths of rates or balance-sheet policies, like Quantitative Easing, as their policy. In this paper, I extend the household's partially backward-looking expectation formation from McMahon & Rholes (2023) in a 3-equation New Keynesian model to incorporate a zero-lower bound constraint. McMahon & Rholes (2023) motivate this boundedly-rational setup based on their experimental finding that subjects' inflation forecasts depend on the central bank's past forecast errors. This boundedly-rational household set-up allows me to study the role of central bank credibility, expressed here as an exogenous parameter, in influencing the welfare cost of the ZLB. I study how the extent to which agent's expectations are anchored crucially affect whether the ZLB is costlier than in the rational expectations case. Further, this essay shows that forward guidance, implemented as the central bank's announcements of deviations from forecast, is less effective in increasing output in the present period.

1.1 Related Literature

Galí (2018) discusses how a standard New Keynesian model with an interest rate rule tracking the natural rate is pushed into a persistent recession at the zero-lower bound. When unable to track a negative real rate, both inflation and output are below their steady state levels in a discretionary central bank framework, with the size of the drop depending on the expected duration of the ZLB episode. It constrains the central bank in using its policy instrument

to counter shocks that require reducing rates, putting the economy at the risk of a deep recession.

Eggertsson & Woodford (2003*b*) study the ZLB, inspired by the then recent situation in Japan, with their central bank's policy rate being within 50 basis points of 0 since October 1995. Through Eggertsson & Woodford (2003*a*), they discuss alternative optimal policies beyond inflation targeting and their implementation, concluding that a history-dependent policy by the bank to credibly commit to keeping nominal rates low for a few periods after the natural rate returns above the ZLB performs better than expansions of the monetary base. By committing to higher average inflation after the economy exits the ZLB, it offsets deflationary pressures in the present period through the expectations channel, bringing the economy out of a recession. Woodford (2013) discusses different ways in which this idea of forward guidance has been used by Central banks around the world to tackle the liquidity trap that arose out of the Financial Crisis in 2008. Del Negro et al. (2012) investigate forward guidance policies by the US Federal Reserve and find a crucial link between the policy's quantitative impact, its type and exact content in the forward guidance announcement, suggesting an important role for central bank communication.

The forward guidance puzzle emerges out of theoretical analyses of forward guidance in rational expectations NK models. Galí (2018) explains it as independence of the policy's enactment period with the magnitude of its impact on output; a rate cut announced to be implemented many periods into the future is as effective as an equivalent rate cut to be enacted the next period. Del Negro et al. (2012) provide an early theoretical exposition for it; showing that a standard medium-term DSGE model grossly overestimates its impact due to no discounting of future interest rates changes in the dynamic IS curve.

Most papers attempting to find a solution to this puzzle rely on models with expectations over future interest rate changes that are progressively more discounted with time horizon. McKay et al. (2016) attempt this with agents facing uninsurable income risk and borrowing constraints, while Farhi & Werning (2019) combine this assumption of incomplete markets with bounded

rationality of agents in the form of 'level-k thinking'.

Based on micro-foundations developed in Gabaix (2014) and Gabaix (2019), Gabaix (2020) resolves the forward guidance puzzle with myopic agents, who cognitively discount expectations over future variables such that future interest changes have a geometrically declining effect on variables in the present with every incremental period. Goy et al. (2022) study forward guidance in a world with heterogeneous agent beliefs, partially formed using a forecast from lagged variables, demonstrating the effectiveness of forward guidance in a liquidity trap.

By analysing the model's response to shocks, this essay contributes to the increasingly relevant work in Behavioural New Keynesian models. It also add to the literature on solutions to the forward guidance puzzle, demonstrating that this boundedly-rational expectations framework implies that output gap expectation in the dynamic IS curve is discounted, i.e. the household is not perfectly forward-looking. The essay relates to literature on central bank communication, particularly its role in making policies at the ZLB effective. Haldane & McMahon (2018) and McMahon & Rholes (2023) connect central bank communication with its credibility and this essay's findings demonstrate how credibility is essential for communication-based policies to be successful.

Furthermore, Gabaix (2020) uses attenuation of forward expectations to show that the economy is stable at the ZLB and it is less costly. McKay et al. (2017) show this result with a discounted Euler equation, with both papers supporting empirical evidence from Japan and the US in Cochrane (2018). Diving deeper, Hommes & Lustenhouwer (2019) show that lack of central bank credibility can cause deflationary spirals at the ZLB. In this essay, I provide more evidence supporting this claim by analysing how the agent's reaction to past forecast errors impacts welfare cost of the ZLB. With parameter values supporting unanchored household expectations, I find that the cost of the ZLB is higher than in the rational case.

The rest of the paper is organised as follows: Section 2 contains the model and theoretical justification for the results. Section 3 explores parameter calibration and the simulation methodology. Section 4 contains results from model simulations along with a discussion, and Section 5 concludes.

2 Model

2.1 Standard New Keynesian Model

I adopt the baseline model from Galí (2015), modifying it with alternative expectations formation from McMahon & Rholes (2023). This model is log-linearised around a steady state with perfect foresight and zero inflation. In the model, log-deviations of inflation and output gap are related according to the New Keynesian Phillips curve:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa y_t + u_t \tag{2.1}$$

where u_t denotes a cost-push shock to inflation at time t and β is the discount factor. The slope of the Phillips curve, κ, is given by:

$$\kappa = \lambda(\sigma + \frac{\phi + \alpha}{1 - \alpha}) \tag{2.2}$$

where $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}\Theta$ and $\Theta = \frac{1-\alpha}{1-\alpha+\alpha\epsilon}$

κ is inversely related to firm-level real rigidities through the price elasticity of demand faced by firms producing differentiated goods in the monopolistic competition market structure (ϵ) and the capital share in the firm's production technology (α), assuming capital is fixed at firm level.

κ also depends on aggregate-level real rigidities, which are inversely related to σ, the intertemporal elasticity of substitution, and on ϕ, which is the inverse of the Frisch elasticity of labour supply.

λ is inversely related to degree of price rigidity (nominal rigidities), and depends on the Calvo parameter of price stickiness θ. In Calvo pricing, θ measures the probability that an individual firm cannot adjust its good's price in a given period.

Following Galí (2015), the relation between output gap (in log-linear terms) to deviations of the real interest rate from the natural rate of interest (r_t^n) is given by the log-linearised dynamic IS equation:

$$y_t = \mathbb{E}_t[y_{t+1}] - \frac{1}{\sigma}(i_t - \mathbb{E}_t[\pi_{t+1}] - r_t^n) + g_t \tag{2.3}$$

where g_t represents an IS curve shock, hereafter interpreted as a temporary demand shock. Note that I use the Fisher equation here in rewriting real interest rate in terms of the nominal interest rate and expected inflation in the next period.

$$r_t = i_t - \mathbb{E}_t[\pi_{t+1}] \tag{2.4}$$

I close the model by assuming that the Central Bank sets nominal interest rates by tracking the natural rate of interest and reacting to deviations from the targets of zero inflation and zero output gap up to an error term, expressed as a log-linearised Taylor rule:

$$i_t^{Taylor} = r_t^n + \phi_\pi \pi_t + \phi_y y_t + v_t \tag{2.5}$$

where the error term v_t is a monetary policy shock, and the natural rate of interest is given by:

$$r_t^n = \rho + \sigma \psi_{ya}(\mathbb{E}_t[a_{t+1}] - a_t) \tag{2.6}$$

where a_t is firm-level productivity. The composite parameter ψ_{ya} is given by

$$\psi_{ya} = \frac{1+\phi}{\sigma(1-\alpha)+\phi+\alpha} \tag{2.7}$$

I impose a zero-lower bound constraint motivated by zero interest from cash and zero cost of storing money in the form of cash.

$$i_t = max\{0, i_t^{Taylor}\} \tag{2.8}$$

2.2 Alternative Expectations Formation

2.2.1 Household Prior Expectation

Most basic New Keynesian models assume that agents form rational expectations, thus, communication cannot play a role in these models as agents have full information on all variables and can perfectly process said information in forming expectations over all future period variable values. Deviating from this benchmark, I follow the model from McMahon & Rholes (2023), assuming a representative Bayesian household that forms a prior expectation of X_t as a composite of forward-looking rational expectations and a backward-looking component. Here, $X_t \in \{\pi_t, y_t\}$

Specifically, let's say that $\xi \in [0, 1]$ proportion of the household's expectation is rational,

$$\mathbb{E}^{HH}_{t,RE}[X_{t+1}] = X^{RE}_{t+1} \tag{2.9}$$

and the remaining $1 - \xi$ forms backward-looking expectation:

$$\mathbb{E}^{HH}_{t,BR}[X_{t+1}] = \mathbb{E}_{t-2}[X_{t-1}] + \eta_X(X_{t-1} - \mathbb{E}_{t-2}[X_{t-1}]) \tag{2.10}$$

Here, the representative agent adjusts its expectation by some weight on past forecast errors: deviations of its ex-post expectation from the realized inflation in the previous period. $\eta_X \in [0, 1]$ represents the extent to which the agent weighs these deviations from the forecast. Thus, the $\eta_X = 1$ benchmark represents an agent that naively updates its backward looking component with previous period inflation, while $\eta_X = 0$ means that the backward-looking component is completely anchored to $\mathbb{E}_{t-2}[X_{t-1}]$.

The agent's composite prior is given by:

$$\mathbb{E}^{HH}_{t,prior}[X_{t+1}] = \xi\mathbb{E}^{HH}_{t,RE}[X_{t+1}] + (1 - \xi)\mathbb{E}^{HH}_{t,BR}[X_{t+1}] + e^{X}_{t,HH} \tag{2.11}$$

where $e^{X}_{t,HH}$ is a Gaussian noise component.

2.2.2 Central Bank Expectation

As Goy et al. (2022) discuss, we can expect that Central banks have much better tools, data and expertise to reliably forecast expectations. I assume that the bank publishes model consistent rational expectation of future inflation, but with a Gaussian noise component $e^{X}_{t,CB}$:

$$\mathbb{E}^{CB}_{t}[X_{t+1}] = X^{RE}_{t+1} + e^{X}_{t,CB} \tag{2.12}$$

2.2.3 Exogenous Credibility

The possibility of shocks to household prior expectation and the central bank's forecast means that the household cannot fully rely on the central bank's published forecast. I follow McMahon & Rholes (2023) in introducing a credibility parameter, $\Gamma \in [0, 1]$, to represent the extent to which the household encompasses the bank's forecast in its posterior. Using Bayes' Theorem, the household's posterior is:

$$\mathbb{E}^{HH}_{t,post}[X_{t+1}|\mathbb{E}^{CB}_{t}[X_{t+1}]] = (1-\Gamma)\mathbb{E}^{HH}_{t,prior}[X_{t+1}] + \Gamma\mathbb{E}^{CB}_{t}[X_{t+1}] \tag{2.13}$$

In this framework henceforth, I assume that the central bank commits to its specified policy rule and other agents in the economy know and believe that the bank won't change the parameters of the policy rule.

2.3 Impact on the Economy at the Zero Lower Bound

2.3.1 Rational Expectations

Multiple previous studies analysing variants of rational expectations NK model, such as Eggertsson & Woodford (2003*b*), describe how the central bank is constrained in its use of policy to respond to shocks. Mertens &

Williams (2019) describes how under discretionary monetary policy, taking the form of inflation targeting, inflation and output gap are constant and are anchored below their steady-state levels at the ZLB.

Assume that the ZLB binds between periods $t = 0$ and $t = T$, where T is unknown to the agents. The natural rate goes to $\tilde{r} < 0$, meaning that the nominal rate cannot track it anymore. Denoting these anchored values as $X^{RE,ZLB}$, both $\pi^{ZLB,RE}$ and $y^{ZLB,RE}$ are less than their non-ZLB steady state levels of 0. The central bank cannot react to any shock that requires reducing nominal interest rate and such a shock drives the economy to an equilibrium with even lower inflation and output gap levels.

2.3.2 Partially Backward-looking Expectations

With the behavioural expectations assumption, when Γ and ξ are less than 1, the economy at ZLB won't be at a constant inflation-output equilibrium anymore as household expectations evolve with forecast errors in inflation and output gap.

At the moment the economy enters the ZLB ($t = 0$), from the NK Phillips curve and IS curve without shocks:

$$\pi_0 = \beta \mathbb{E}_0[\pi_1] + \kappa y_0 \tag{2.14}$$

$$y_0 = \mathbb{E}_0[y_1] - \frac{1}{\sigma}(-\mathbb{E}_0[\pi_1] - \tilde{r}) \tag{2.15}$$

As the agent does not know whether they exit the ZLB next period, expectations do not fully adjust: $|\mathbb{E}_0[\pi_1]| < |\pi_0|$ and $|\mathbb{E}_0[y_1]| < |y_0|$ and hence from the above equations, $\pi_0 < 0, y_0 < 0$. For the next period, $1 - \xi$ proportion of backward-looking households use this value to update their expectation for next period inflation, noting that $\mathbb{E}_{-1}[\pi_0] = 0$:

$$\mathbb{E}^{HH}_{1,BR}[\pi_2] = \eta_\pi \pi_0 \tag{2.16}$$

Assuming no shocks to household's inflation expectation, with $E^{HH}_{1,RE}[\pi_2] = \pi_2^{RE}$ the household updates it's prior to:

$$\mathbb{E}^{HH}_{1,prior}[\pi_2] = \xi\pi_2^{RE} + (1-\xi)\eta_\pi\pi_0 \tag{2.17}$$

From central bank's inflation forecast in 2.12, I write $\mathbb{E}_1^{CB}[\pi_2] = \pi_2^{RE} + e^{\pi}_{2,CB}$, and the posterior is updated to:

$$\mathbb{E}_1[\pi_2] = (1-\Gamma)((1-\xi)\eta_\pi\pi_0 + \xi\pi_2^{RE}) + \Gamma\pi_2^{RE} + \Gamma e^{\pi}_{2,CB} \tag{2.18}$$

Noting that $i_1 = 0$, this updated expectation feeds into next period's output gap as:

$$y_1 = \mathbb{E}_1[y_2] - \frac{1}{\sigma}(-(1-\Gamma)(1-\xi)\eta_\pi\pi_0 - ((1-\Gamma)\xi + \Gamma)\pi_2^{RE} + \Gamma e^{\pi}_{2,CB} - \tilde{r}) \tag{2.19}$$

Assuming the Central bank's forecast is not being updated ($e^{\pi}_{2,CB} = 0$), the household accounts for the deviation of the realized lower inflation from higher forecast inflation value and lowers its expectations of next period's inflation, creating a downward pressure on output gap. This downward pressure is balanced to some extent by the $\Gamma + (1-\Gamma)\xi$ coefficient on model-consistent next period inflation, which from section 2.3.1 is expected to be negative. This is a reflection of the fact that the agents are not perfectly forward looking; I elaborate upon this concept in 2.4.2.

Using an analogous derivation for output gap, assuming no shocks to household's expectation of the output gap, I find that:

$$\mathbb{E}_1[y_2] = (1-\Gamma)((1-\xi)\eta_y y_0 + \xi y_2^{RE}) + \Gamma y_2^{RE} + \Gamma e^{y}_{2,CB} \tag{2.20}$$

As with inflation, we see a downward pressure on expectations due to the household's adjustment of output expectations, which is balanced to some extent by the < 1 coefficient on y_2^{RE}.

On plugging in 2.19 in 2.1:

$$\pi_1 = \beta\mathbb{E}_1[\pi_2] + \kappa(\mathbb{E}_1[y_2] - \frac{1}{\sigma}(-(1-\Gamma)(1-\xi)\eta_\pi\pi_0 \\ - ((1-\Gamma)\xi + \Gamma)\pi_2^{RE} + \Gamma e_{2,CB}^{\pi} - \tilde{r})) \quad (2.21)$$

where the entire second term after κ is negative. Without any correction of forecasts by the central bank, 2.18, 2.19, 2.20 and 2.21 suggest a cascading effect on inflation and output gap at the ZLB due to the household's downward correction of expectations of each variable, which in turn further lowers the other variable. Compared to the rational expectations case, the size of the recession would depend on parameter values, as we cannot rule out the correction in the downward pressure by household's discounting of forward-looking expectation.

The values of η_π and η_y play a crucial role in quantifying downward pressures on π and y due to their role in determining expectation correction via past forecast errors. $\eta_\pi = 0$ and $\eta_y = 0$ correspond to perfectly anchored expectations: $\mathbb{E}_{t,BR}^{HH}[X_{t+1}] = \mathbb{E}_{t-2}[X_{t-1}]$ for $X \in \{\pi, y\}$. As inferred from 2.18 and 2.20 with $\eta_X = 0$, expectations for both variables remain anchored at the rational expectations benchmark. We should expect to see the ZLB be less costly than in the perfectly rational agents case, due to the imperfect forward looking behaviour, represented by the composite parameter $(1-\Gamma)\xi+\Gamma$ being < 1.

On the contrary, with $\eta_X = 1$, household's expectation is naively updated to the previous period's value for that variable, taking no account of how the posterior expectations themselves are being updated. Starting with a negative initial value, we should expect to see the highest downward pressure on inflation and output.

Not surprisingly, η_X does not play a role in determining the forward looking component's impact on inflation and output in the present period. I explore this idea further in the next section.

2.4 Analysis of Forward Guidance

2.4.1 Rational Expectations

To assess the effectiveness of forward guidance, I start with the dynamic IS equation in 2.3 without the shock. Assuming that the agents (both firms and households) are fully rational, the Expectation operator is rational expectations, and we can replace the expectation of next period output gap with its model consistent RE value and use the Fisher equation 2.4 to re-write it as:

$$y_t = \mathbb{E}_{t,RE}[y_{t+1}] - \frac{1}{\sigma}(r_t - r_t^n) \tag{2.22}$$

Substituting successive values of y_{t+j} and iterating it forward, we get:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{T}\mathbb{E}_{t,RE}[(r_{t+j} - r_{t+j}^n) + \mathbb{E}_{t,RE}[y_{t+T}] \tag{2.23}$$

With $T \to \infty$, applying the transversality condition: $\lim_{T\to\infty}\mathbb{E}_{t,RE}[y_{t+T}] = 0$, we get:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}\mathbb{E}_{t,RE}[(r_{t+j} - r_{t+j}^n)] \tag{2.24}$$

Using the Fisher equation, assuming no shocks to the natural rate:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}(\mathbb{E}_{t,RE}[i_{t+j} - \pi_{t+j+1}] - r_{t+j}^n) \tag{2.25}$$

To simplify, I assume a simpler version of the Taylor rule from 2.5 with $\phi_\pi > 1$, $\phi_y = 0$, and no shock; these parameter values are sufficient for a determinate equilibrium according to Galí (2015). On simplifying:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}(\mathbb{E}_{t,RE}[\phi_\pi\pi_{t+j} - \pi_{t+j+1}] - r_{t+j}^n) \tag{2.26}$$

From 2.13 combined with 2.12, with parameter values consistent with fully rational expectations, $\Gamma = 1$, we get that:

$$\mathbb{E}_{t,RE}[\pi_{t+j}] = \mathbb{E}_{t,CB}[\pi_{t+j}] = \pi^{RE}_{t+j} + e^{\pi}_{t+j,CB} \tag{2.27}$$

where $e^{\pi}_{t+j,CB}$ are shocks to central's bank inflation expectations. Unlike Del Negro et al. (2012), where the authors use shocks to the short-term interest rate (i.e. monetary policy shocks) as a realisation of forward guidance, this expectations formation mechanism suggests that announced deviations from the central bank's benchmark forecast will show up as shocks to the central bank's expectations. Thus, I assume that forward guidance in this model is a series of shocks to $e^{\pi}_{t+j,CB}$ and $e^{y}_{t+j,CB}$ for some series of $j > 0$ that the agent learns at time t.

I rewrite this equation with model-consistent future inflation terms:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}(\phi_\pi \pi^{RE}_{t+j} + \phi_\pi e^{\pi}_{t+j,CB} - \pi^{RE}_{t+j+1} - e^{\pi}_{t+j+1,CB} - r^{n}_{t+j}) \tag{2.28}$$

From 2.28, since $e^{\pi}_{t+j,CB}$ in each future period has the same coefficient, the bank's announcement of a fall in inflation through negative future expectation shocks at any period in the future has the same expansionary effect on output gap as an announcement of a negative expectation shock in the next period. Effectively, the central bank can achieve a very high $\Delta y_t > 0$ through announcement of forecast changes in the far future.

For completeness, I analyse how inflation in this model responds to forward guidance. Iterating 2.1 forward and imposing the terminal condition $\lim_{T\to\infty} \mathbb{E}_{t,RE}[\pi_{t+T}] = 0$:

$$\pi_t = \kappa \sum_{j=0}^{\infty} \beta^j \mathbb{E}_{t,RE}[y_{t+j}] \tag{2.29}$$

Abstracting away from shocks to central bank's output expectations, $e^{y}_{t+j,CB} = 0$, a large enough Δy^{RE}_t is sufficient to overcome the negative inflation expectation shocks and induce a positive deviation in inflation in the present

period: $\Delta\pi_t^{RE} > 0$

2.4.2 Partially Backward-looking Expectations

Now, I show that strength of forward guidance in the behavioural model is smaller than in the rational case by deriving a geometrically decreasing slope coefficient on successive real rates in the IS curve.

First, re-writing the posterior expectation of future macroeconomic variables in terms of past realized and forecast terms, and expected future values under rational expectations by exploiting my assumptions relating the boundedly-rational expectation operator with rational expectations.

Plugging in 2.9 and 2.10 in 2.11 to express the household prior in RE terms (assume $\eta_\pi = \eta_y = \eta_X$):

$$\begin{aligned}\mathbb{E}^{HH}_{t,prior}[X_{t+1}] = \xi\mathbb{E}^{HH}_{t,RE}[X_{t+1}] + (1-\xi)\mathbb{E}_{t-2}[X_{t-1}] \\ + (1-\xi)\eta_X X_{t-1} - (1-\xi)\eta_X\mathbb{E}_{t-2}[X_{t-1}] + e^X_{t,HH}\end{aligned} \tag{2.30}$$

Substituting 2.12 and 2.30 in 2.13, using $\mathbb{E}^{HH}_{t,post}[X_{t+1}|\mathbb{E}^{CB}_t[X_{t+1}]] = \mathbb{E}_t[X_{t+1}]$ and collecting the $X^{RE}_{t+1} = E^{HH}_{t+1,RE}[X_{t+1}]$ terms:

$$\begin{aligned}\mathbb{E}_t[X_{t+1}] = (\Gamma + (1-\Gamma)\xi)X^{RE}_{t+1} + (1-\Gamma)((1-\xi)\mathbb{E}_{t-2}[X_{t-1}]+ \\ (1-\xi)\eta_X X_{t-1} - (1-\xi)\eta_X\mathbb{E}_{t-2}[X_{t-1}]) + \Gamma e^X_{t,CB} + (1-\Gamma)e^X_{t,HH}\end{aligned} \tag{2.31}$$

The composite coefficient $\Gamma + (1-\Gamma)\xi$ on X^{RE}_{t+1} is ≤ 1, with equality when $\Gamma = 1$ or $\xi = 1$. Furthermore, we can see here that η_X, the degree to which households weigh past forecast errors, does not impact this coefficient on forward looking RE inflation.

Substituting 2.31 for y_t in the dynamic IS equation, 2.3, gives us the modified IS equation:

$$y_t = (\Gamma + (1-\Gamma)\xi)\mathbb{E}_{t,RE}[y_{t+1}] - \frac{1}{\sigma}(r_t - r_t^n)$$
$$+ (1-\Gamma)((1-\xi)\mathbb{E}_{t-2}[y_{t-1}] + (1-\xi)\eta_X y_{t-1} - (1-\xi)\eta_X \mathbb{E}_{t-2}[y_{t-1}]) + \tilde{g}_t \tag{2.32}$$

where $\tilde{g}_t = \Gamma e^y_{t,CB} + (1-\Gamma)e^y_{t,HH} + g_t$ is the composite demand shock term. Forecast errors in the backward-looking expectation component, weighted with η_X, do not form a part of the mechanism through which future expectations affect present output gap. For this reason, I ignore these terms in my analysis henceforth in this section. Specifically, collecting these terms along with the composite shock in a new variable
$\mu_t := (1-\Gamma)((1-\xi)\mathbb{E}_{t-2}[y_{t-1}] + (1-\xi)\eta_X y_{t-1} - (1-\xi)\eta_X \mathbb{E}_{t-2}[y_{t-1}]) + \tilde{g}_t$
and assuming future shocks terms are bounded, iterating forward 2.32 like in 2.23 and with the appropriate transversality condition, we get:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}(\Gamma+(1-\Gamma)\xi)^j \mathbb{E}_{t,RE}[(r_{t+j} - r^n_{t+j})] + \sum_{j=0}^{\infty}(\Gamma+(1-\Gamma)\xi)^j \mu_{t+j} \tag{2.33}$$

Substituting for r_{t+j} using the Fisher equation and reusing the simpler version of the Taylor rule described earlier: $\phi_\pi > 1$ and $\phi_y = 0$:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}(\Gamma + (1-\Gamma)\xi)^j \mathbb{E}_{t,RE}[(\phi_\pi \pi_{t+j} - \mathbb{E}_{t+j}[\pi_{t+j+1}]] - r^n_{t+j})$$
$$+ \sum_{j=0}^{\infty}(\Gamma + (1-\Gamma)\xi)^j \mu_{t+j} \tag{2.34}$$

Rewriting 2.34 in terms of the Central bank's noisy forecast by replacing the posterior rational expectation over π_{t+j} in terms of its Bayesian updating from 2.13, with $\Gamma = 1$ (as rational expectations fully incorporates the central bank's forecast) and substituting from the central bank's expectation in 2.12, we get:

$$y_t = -\frac{1}{\sigma}\sum_{j=0}^{\infty}(\Gamma + (1-\Gamma)\xi)^j[(\phi_\pi \pi_{t+j}^{RE} + \phi_\pi e_{t,CB}^{\pi} - \mathbb{E}_{t+j}[\pi_{t+j+1}]] - r_{t+j}^{n}) + \sum_{j=0}^{\infty}(\Gamma + (1-\Gamma)\xi)^j \mu_{t+j} \quad (2.35)$$

As in the standard version, we can think of forward guidance in this model as an announcement of decline in future inflation by the central bank, which for the household presents as unanticipated expectation shocks in the period the announcement happens, and anticipated thereafter. In the model, it appears as a series of shocks $e_{t+j,CB}^{\pi}$ announced at time t as described in 2.4.1. From 2.35, with $\lim_{j\to\infty}(\Gamma + (1-\Gamma)\xi)^j \to 0$, this sequence of shocks has a geometrically declining effect on output in present period compared to the standard model in 2.28.

Replicating the steps in deriving 2.29 but with a general expectations operator, we get:

$$\pi_t = \kappa \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t[y_{t+j}] \quad (2.36)$$

And with y_t lower than in the rational expectations case, π_t cannot rise as much with the same series of anticipated negative expectation shocks compared to forward guidance with rational expectations: $\Delta y_t^{RE} > \Delta y_t$ and $\Delta \pi_t^{RE} > \Delta \pi_t$

To summarize, I would expect forward guidance to be weaker as a policy instrument available to an imperfectly credible central bank, with the strength of forward guidance declining with falling credibility. Since the composite term $\Gamma + (1-\Gamma)\xi$ determines strength of forward guidance, a credible central bank can get very strong forward guidance based output boosts even when the agents are not fully forward looking ($\xi < 1$).

3 Simulation

3.1 Calibration

The parameters related to the standard NK are calibrated based on Galí (2015), with each period corresponding to a quarter. A description of these parameters and the values assigned is given below.

Symbol	**Description**	**Input Value**
β	Household discount factor	0.99
σ	Intertemporal elasticity of substitution	1
ρ	Steady state natural rate of interest	0.0101
ϕ	Inverse of Frisch elasticity of labour supply	5
$1-\alpha$	Labour share in Production function	0.75
ϵ	Demand elasticity	9
θ	Calvo parameter/ Survival rate of prices	0.75
ϕ_π	Coefficient on inflation in Taylor Rule	1.5
ϕ_y	Coefficient on output gap in Taylor Rule	0.5/4

Table 3.1: Standard Input Parameter values

These parameter values imply the following composite parameters:

Symbol	**Description**	**Input Value**
κ	Slope of Phillips curve	0.172
ψ_{ya}	Slope coefficient on change in productivity in natural rate equation divided by σ	1

Table 3.2: Composite Parameter values

I base the calibration of parameters in the behavioural expectations formation on papers that estimate the model in Gabaix (2020). I use combinations of parameters (Γ, ξ) such that the coefficient in front of the interest rate term in 2.35 matches the point estimate for cognitive discounting term on $\mathbb{E}_t[y_{t+1}]$ in the IS curve in this paper, that also appears as the geometrically decreasing coefficient on future interest rates on forward iterating the behavioural IS equation. Andrade et al. (2019) obtain an MLE estimate of 0.6799, and a continuous updating GMM estimate of 0.393 based on the methodology

from Mavroeidis et al. (2014). Afsar et al. (2024) obtain an estimate of 0.46 in a Behavioural NK model augmented with backward-looking households through habit formation and price indexation, while Ilabaca et al. (2020) obtain an estimate of 0.7059 for a model with determinate equilibria. More recently, Ilabaca & Meggiorini (2023), using values from the Ilabaca et al. (2020), obtain an estimate of 0.42 for cognitive discounting in a model with non-zero trend inflation.

Based on the estimates above, I split my parameter values into 4 sets for the simulation:

1. Rational expectations benchmark - either $\Gamma = 1$ or $\xi = 1$, for any η_X
2. Moderate cognitive discounting equivalent: $\Gamma + (1 - \Gamma)\xi = 0.7$. This benchmark is obtained on setting $(\Gamma, \xi) = (0.6, 0.25)$ and $(\Gamma, \xi) = (0.2, 0.625)$
3. High cognitive discounting equivalent: $\Gamma + (1 - \Gamma)\xi = 0.4$. Obtained on setting $(\Gamma, \xi) = (0.2, 0.25)$
4. Zero parameter value: Set either $\Gamma = 0$ or $\xi = 0$, with the other parameter at its high cognitive discounting benchmark value: $\Gamma + (1 - \Gamma)\xi = 0.25, 0.2$

In the later three sets, η_X is set at 0.5. To analyse differences in impulse responses on changing the reaction to past forecast errors, I vary η_X from 1 (naively backward-looking expectations) to 0.5 to 0 (fully anchored expectations) with the other 2 parameters at the high cognitive discounting values.

3.2 Setup

3.2.1 Stochastic Simulation of Behavioural NK Model

In a counterfactual exercise, I obtain impulse responses of output gap, inflation and nominal interest rate by alternatively varying credibility (Γ) and

share of forward-looking expectation component in household prior (ξ), keeping the other constant to achieve the benchmark values from the calibration above.

This model has 2 AR(1) shocks defined as:

- Monetary policy shock: $v_t = \rho_v v_{t-1} + \epsilon_t^v$
- Demand shock (IS curve shock): $g_t = \rho_g g_{t-1} + \epsilon_t^g$

Where each of the error terms is distributed i.i.d. $\mathcal{N}(0, 1)$. Shock persistence values are set as: $\rho_v = 0.5, \rho_g = 0.5$.

3.2.2 Impact of the ZLB

To obtain the economy's response to unanticipated productivity shocks, I implement a zero-lower bound on the nominal interest using the OccBin toolkit in Dynare, developed in Guerrieri & Iacoviello (2015). Through this toolkit, we obtain a piecewise linear simulation for a model with occasionally binding constraints via a first-order perturbation approach, making it possible to analyse the response of model variables to unanticipated shocks under uncertainty. I introduce unanticipated productivity shocks of 1% for 5 periods with zero persistence $\rho_a = 0$. Since the agent does not see how many periods the shock is for, it drives the natural rate below zero. For the behavioural expectations case, the simulation is run for parameter values of $\Gamma = 0.7$ and $\xi = 0.7$, as the code does not converge for lower values (which is expected, since the model is log-linearised). Additionally, I compare results between naively-backward looking expectations ($\eta_X = 1$), partially backward looking ($\eta_X = 0.5$) and fully anchored expectations ($\eta_X = 0$) to analyse when the binding constraint is the costliest.

Due to non-convergence with lower credibility and behavioural parameter values, I setup another ZLB simulation in a perfect foresight setting: agents observe all current and future shocks at the time of model initialisation, and solve the model numerically conditional on these shocks. As such, it is not possible to observe consumer behaviour pertaining to uncertainty of future

shocks, such as precautionary savings. The ZLB constraint is implemented and solved with a Levenberg-Marquardt Mixed Complementarity Problem (LMMCP) solver in Dynare, adapted from Kanzow & Petra (2004). I expand upon the setup and results in Appendix B

3.2.3 Effectiveness of Forward Guidance

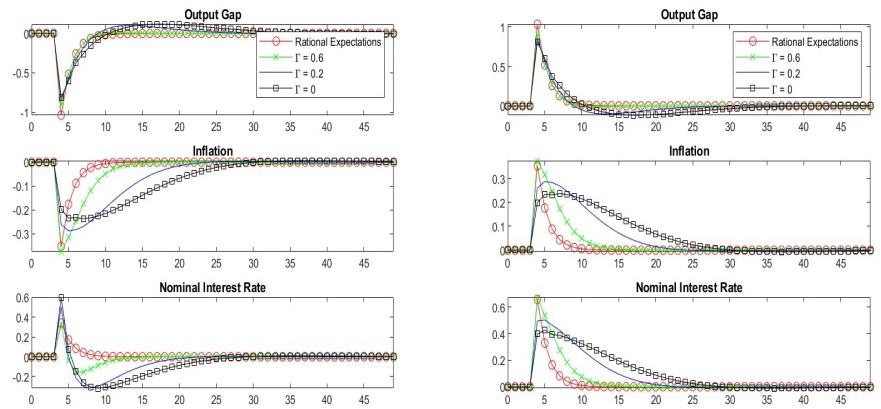
I return to the perfect foresight version of the model to implement forward guidance, due to the ease of implementing anticipated shocks to expectations in this setup. As the findings related to forward guidance are equally applicable to a general situation away from ZLB, I shock the model with a large one-period positive monetary policy shock of 5% and persistence $\rho_v = 0.9$. Forward guidance in this economy is an announcement of deflationary shocks by the central bank, which take the form of shocks to central bank expectations, $e^{\pi}_{CB,t}$, in 2.12. The economy faces these shocks from periods 2 to 15 and are of size -0.5%, with no persistence. Effectively, the forward guidance here reveals information about the central bank's forecasts, and thus is closer to what the literature describes as Delphic (see Campbell et al. (2012)). As my aim is to compare the strength of forward guidance between the rational and behavioural case, I use forecasts of negative future inflation, which corresponds to a revelation of low rates, to boost output in the present period. Note that I keep the ZLB constraint in this model (via the mixed-complementarity problem solver LMMCP) to preclude the possibility of negative interest rates after the shock subsides as a reaction to any remaining negative output gap and inflation.

Owing to the perfect foresight nature of the set-up, the agent learns the whole path of shocks at $t = 0$ and can optimize its behaviour in period 0, in a way replicating a real-world forward guidance announcement. I first compare IRFs with and without forward guidance for rational expectations and then follow it up with that comparison for high cognitive discounting equivalent ($\Gamma = 0.2, \xi = 0.25$), moderate cognitive discounting ($\Gamma = 0.6, \xi = 0.25$) and zero credibility ($\Gamma = 0, \xi = 0.25$).

4 Results and Discussion

4.1 Stochastic Simulations without the ZLB

In Fig. 4.1a, a 1% monetary policy shock leads to a fall in output gap and inflation in both rational expectations and behavioural models. While the output gap's response is very similar irrespective of the value of Γ, inflation tends to be more persistent with lower credibility, despite the presence of backward-looking expectation component in both inflation and output gap. This difference in output gap and inflation's response is most prominently visible in the zero credibility case.



(a) 1% Monetary Policy Shock (b) 1% Demand (IS curve) Shock

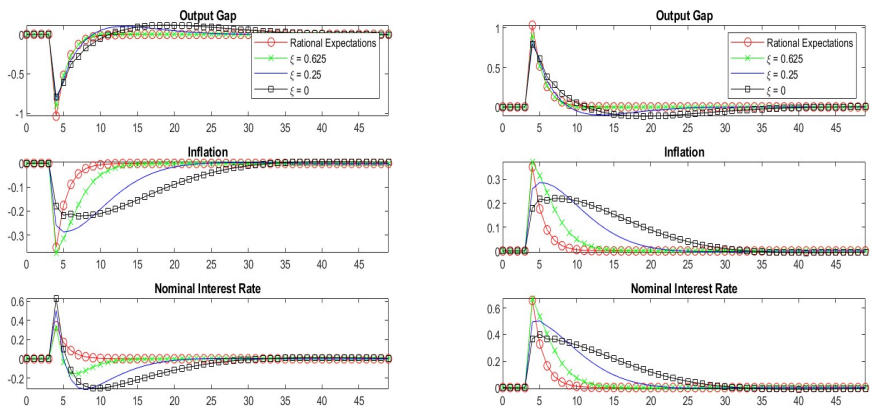
Figure 4.1: Impulse Responses from baseline model, $\xi = 0.25$ and $\eta_X = 0.5$, variation in credibility Γ

Analysis of the model's response to a 1% demand shock, in Fig. 4.1b, is similar to that with an MP shock, as output gap's response does not vary on varying credibility, while inflation becomes more persistent with falling credibility.

Fig. 4.2 repeats the exercise but with Γ constant at 0.2 and variation in the weight on the household's forward looking expectation, ξ. Each of the four benchmarks (RE, high cognitive discounting, low cognitive discounting and zero credibility/ forward looking component) correlate very closely with the

results in Fig. 4.1, supporting the importance of a composite term, such as $\Gamma + (1 - \Gamma)\xi$, in explaining variation in impulse responses between the RE case and the behavioural cases.

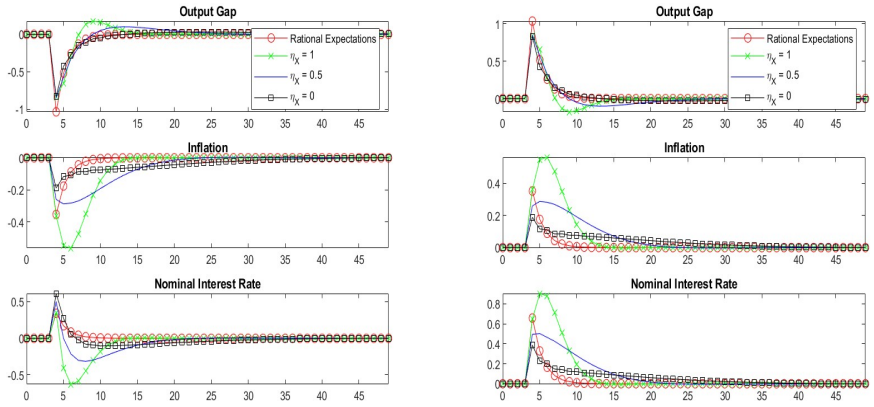
This phenomenon of increasing persistence in inflation and lack thereof in output gap can be explained through the response of the adjusting policy rate. Changes in interest rate are transmitted to output instantaneously through substitution effect seen via the IS curve, 2.3, meaning that output responds faster. Inflation relies on changes in output gap and on next period inflation expectation, so is indirectly affected by rate change. Another way to look at this mechanism is to analyse the response of the variable's expectations relative to the variable, which I present in Appendix A. For both shocks, output gap expectation's response is dampened compared to that of output gap in Fig. A.1 - with the impact response in expectations is half that of output (and its magnitude decreasing with decreasing credibility). This suggests a quick correction in the agent's output expectations due to changes in policy rate. On the other hand, in Fig. A.2, the magnitude of inflation's and its expectations' response is very similar. Fig. A.3 and Fig. A.4 present an analogous analysis with variation in ξ.



(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure 4.2: Impulse Responses from baseline model, credibility $\Gamma = 0.2$ and $\eta_X = 0.5$, variation in forward looking expectation component

In the next part, I ascertain the importance of forecast errors by varying η_X while keeping Γ and ξ fixed. With $\eta_X = 1$, there is amplification in inflation's



(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure 4.3: Impulse Responses from baseline model, credibility $\Gamma = 0.2$ and $\xi = 0.25$, variation in reaction to past forecast errors

response to both a 1% MP shock and a 1% demand shock, which can be corroborated with the fact that the household naively uses its past variable values, π_{t-1} and y_{t-1}, without any weight on the posterior expectation to partially cancel out any deviations from steady state. For $\eta_X < 1$, the term $X_{t-1} - \mathbb{E}_{t-2}[X_{t-1}]$ provides a buffer against this amplification, as the agent's expectations are updated in line with the changing variable values every period.

With $\eta_X = 0$, IRFs very closely resemble the RE case as expectations are completely anchored; the household only uses the forecast $\mathbb{E}_{t-2}[X_{t-1}]$ in the prior. I present the IRFs for expectations with η_X variation in Appendix A in Fig. A.5 and Fig. A.6 respectively. Anchoring of expectations is seen with $\eta_X = 0.5$ and more so with $\eta_X = 0$, with the expectations responding less strongly for these values than in the RE case. Output gap expectations oscillate for high η_X, while inflation expectations get amplified, explaining the trends seen in these two variables when $\eta_X = 1$. Since the magnitude of output gap expectations response is much smaller, its oscillation does not introduce much oscillation in output gap itself. Please refer to Appendix A for more details.

4.2 Impact of the ZLB

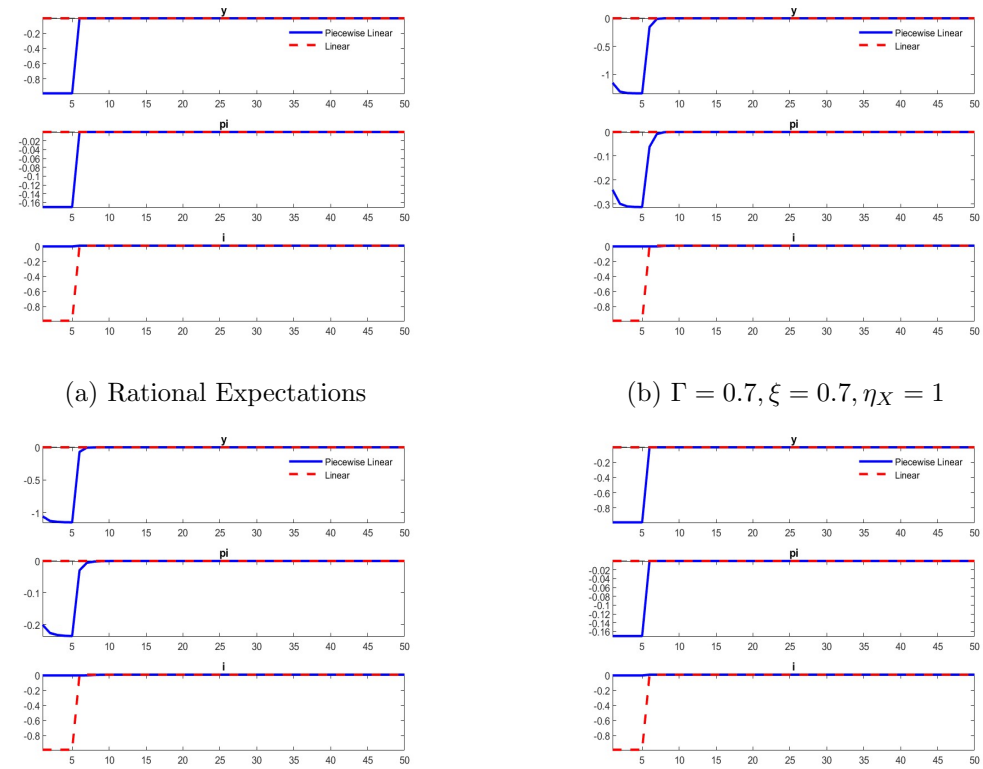
Fig. 4.4 is obtained on running the simulation specified in section 3.2.2. The focus in this figure would entirely be on the piece-wise linear response as the linear response cannot incorporate the non-linear binding constraint. Comparing Fig. 4.4a and Fig. 4.4b, the drop in inflation with $\eta_X = 1$ is almost twice that in the RE case, and the drop in output gap is about 1.2 times. This supports my claim from section 2.3.2 that the ZLB is costlier in a behavioural model with imperfect central bank credibility and naively formed household's backward-looking expectation component. The drag in expectation formation from lowered y_{t-1} and π_{t-1} in the backward-looking component also means that these variables do not reach their minimum values until a few periods into the ZLB episode, which can also be seen when $\eta_X = 0.5$.

With $\eta_X = 0.5$ in Fig. 4.4c, the drop is output gap and inflation is lower than in the $\eta_X = 1$ case but still greater than RE, while the IRFs with $\eta_X = 0$ match the RE case very closely, including in the absence of an initial lag in reaching minima. This supports the claim from 2.3.2 that with anchored expectations, the cost increasing channel in this boundedly-rational model is shutoff, keeping only the cost reducing imperfect forward-looking channel active. Further research should look into obtaining converging simulations with lower Γ and ξ values. I suspect that with anchored expectations, it will find the ZLB to be less costly, when the welfare cost reducing cognitive discounting channel overpowers the cost increasing backward channel discussed in 2.3.2.

I explore results from 3.2.2's Perfect Foresight simulation, where the ZLB is implemented with LMMCP, in Appendix B. This setup allows analysis with lower values of Γ and ξ corresponding to the cognitive discounting benchmarks from 3.1.

4.3 Effectiveness of Forward Guidance

From the left panel of Fig. 4.5, it is clear that forward guidance policy specified in 3.2.3 is strong enough to counteract the effects of a large and highly



(a) Rational Expectations

(b) $\Gamma = 0.7, \xi = 0.7, \eta_X = 1$

(c) $\Gamma = 0.7, \xi = 0.7, \eta_X = 0.5$

(d) $\Gamma = 0.7, \xi = 0.7, \eta_X = 0$

Figure 4.4: Stochastic simulations for model at the ZLB with different extents of reaction to past forecast error

persistent MP shock. It boosts output gap to almost -1% from the counterfactual -2% in the 1st period, at the cost of lower future inflation. The lack of response in interest rate shows that only expectation shocks drive this rise in output.

This phenomenon is absent in the right-side panel, at the high cognitive discounting benchmark. The IRFs with and without forward guidance are almost coinciding, with the small value of $\Gamma+(1-\Gamma)\xi = 0.4$ rendering forward guidance ineffective, supporting the derivation in 2.4.2.

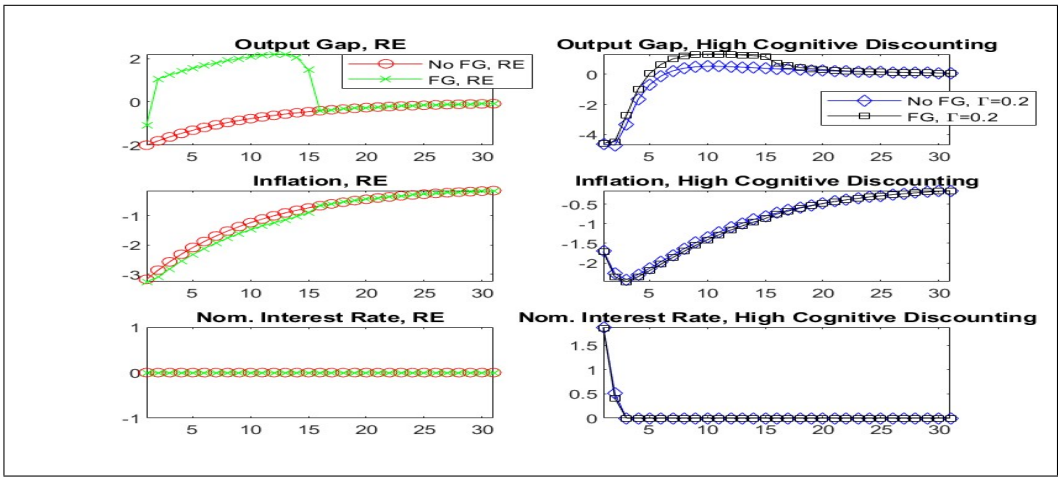


Figure 4.5: Forward Guidance via CB expectations shock in response to MP shock, comparing RE with high cognitive discounting benchmark ($\Gamma = 0.2, \xi = 0.25, \eta_X = 0.5$)

Repeating these simulations for the moderate cognitive discounting benchmark ($\Gamma + (1 - \Gamma)\xi = 0.7$) and zero credibility case in Fig. 4.6 makes the argument clearer. The left panel shows an effect intermediate between the RE and high cognitive discounting case, while in the right panel, with $\Gamma + (1 - \Gamma)\xi = 0.25$, forward guidance's effect on output is so small as to be indiscernible from the plots. This analysis underscores the importance of credibility in central bank communication, suggesting that forward-guidance as implemented by central banks is not as effective as suggested by standard DSGE models due to the banks' imperfect central bank credibility and boundedly-rational agents.

One limitation of perfect foresight here is that the agent know the extent of shocks, in effect, reducing how useful a deviation in forecast is in affecting

output. Ideally, one would evaluate these policies in a stochastic model, where the ZLB binds due to changes in the natural rate. Introduction of uncertainty would mean that commitment by the central bank to changes in reaction function would serve better as forward guidance (see Andrade & Ferroni (2021)). Simulations that model it in a binding ZLB episode can also help inform whether this rationale of Odyssean forward guidance being better than Delphic holds in a model with behavioural agents, particularly accounting for the difficulty in communicating Odyssean policies.

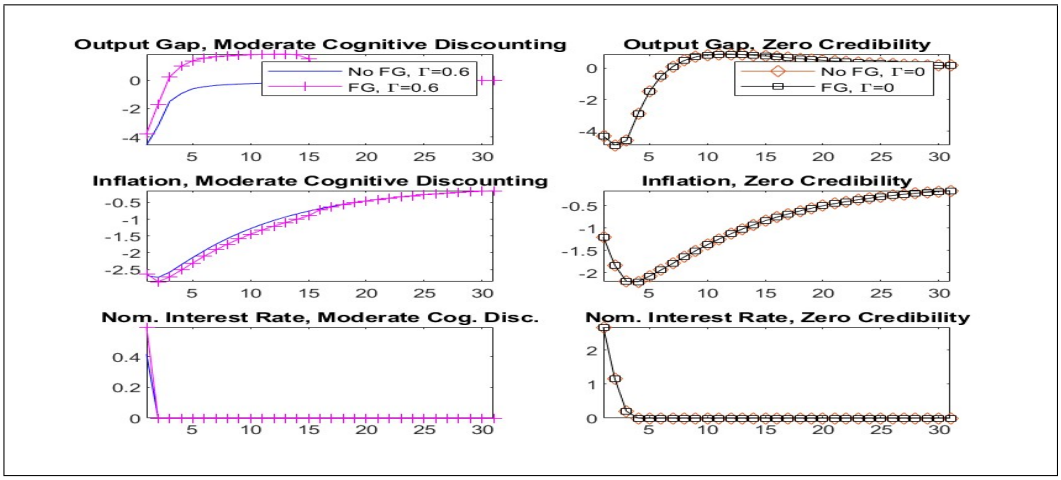


Figure 4.6: Forward Guidance via CB expectations shock in response to MP shock, comparing moderate cognitive discounting benchmark ($\Gamma = 0.6, \xi = 0.25, \eta_X = 0.5$) and zero credibility ($\Gamma = 0, \xi = 0.25, \eta_X = 0.5$)

To justify my claim that η_X value does not affect the power of forward guidance, I repeat this exercise with variation in η_X in Appendix C. Fig. C.1 and Fig. C.2 shows that forward guidance has more or less the same effect at both extremes of η_X, when $\Gamma + (1 - \Gamma)\xi$ is held constant.

5 Conclusion

With boundedly-rational expectations formation, this essay shows that the ZLB may be costlier than in the rational case. Bounded-rationality affects welfare cost through two opposing channels - a forward looking one that reduces the cost and a backward looking one that increases it. Forward guidance is definitively less powerful in the behavioural model, with its strength inversely related to $\Gamma + (1 - \Gamma)\xi$, the coefficient on output expectations in the modified IS curve. It presents an explanation for the forward guidance puzzle and highlights the role of central bank credibility in making forward guidance effective.

IRFs from model simulations show that incorporating boundedly-rational expectations formation introduces persistence and amplification in the response of inflation to monetary policy and demand shocks. The extent of this persistence is inversely related to credibility and extent of forward-looking component in expectations, and amplification is directly related to the weight on the agent's reaction to past forecast errors. Results from stochastic simulations with binding ZLB followed by an implementation of forward guidance in a perfect foresight setting support my theoretical claims about potentially higher welfare costs of the ZLB and the ineffectiveness of forward guidance relative to the standard rational expectations case.

This model does not account for uncertainty when evaluating forward guidance, which will affect its power in boosting the economy. Along with that, future work should look into comparisons of different types of forward guidance policies in their strength and ease of implementation. Further research on this topic should also consider endogenous changes in credibility and weight on past forecast errors, as these terms heavily influence the economy's response at the ZLB. Estimating the model with data will pinpoint these dynamics, helping evaluate how the evolution of credibility and expectation anchoring over time has played a role in the effectiveness of different types of central bank communication.

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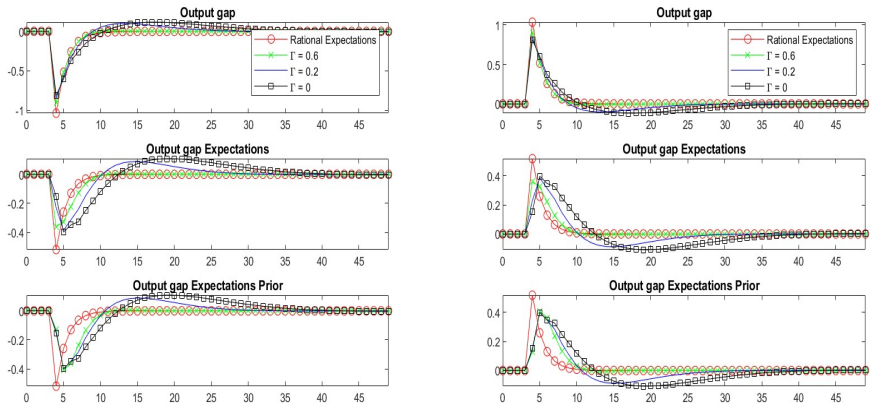
A Appendix: Expectations in Stochastic Simulations

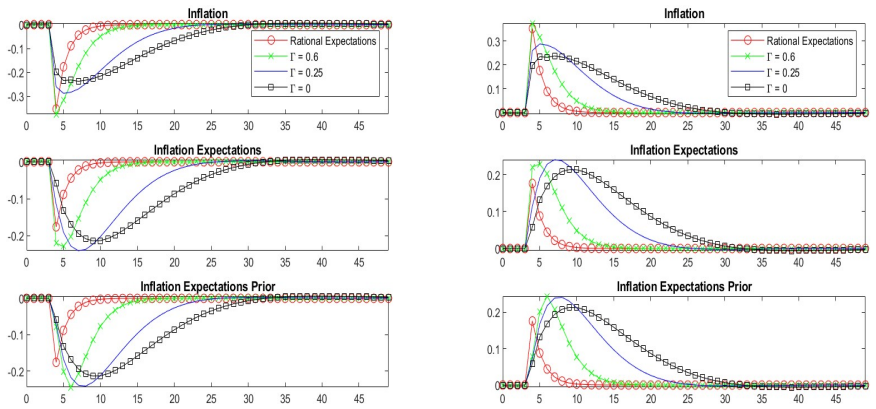
This section reports the IRFs from stochastic simulation of the model without the ZLB for output gap's and inflation's expectations and household expectation prior. We observe the following general trends in these figures:

- The response of inflation expectations is comparable in magnitude to that of inflation itself
- Output gap expectations response is almost half in magnitude of output gap
- Inflation expectations display persistence just like inflation itself: increasing persistence with decreasing credibility and decreasing forward-looking component, but the peak response occurs slightly latter in expectations
- Output gap expectations display some dampened oscillation before reverting to the steady state level; this is not very prominent in output gap itself
- With increasing η_X, inflation expectations response gets amplified, in line with inflation
- Output gap expectations become more oscillatory with increase in η_X, which explains why there is some dampened oscillation of output gap above zero in response to an MP shock (and below zero for the demand shock) in the $\eta_X = 1$ case. Nevertheless, due to the lower magnitude of expectations response, this oscillation is minimal



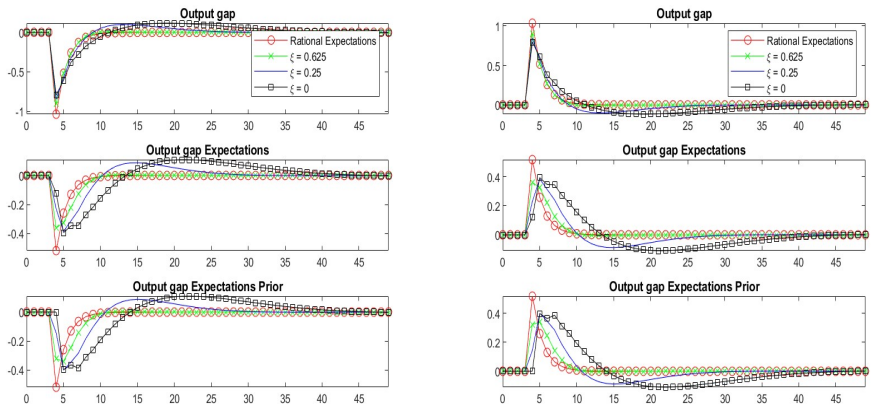
(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure A.1: Impulse Responses for output gap and its expectations from baseline model, $\xi = 0.25$ and $\eta_X = 0.5$, variation in credibility Γ



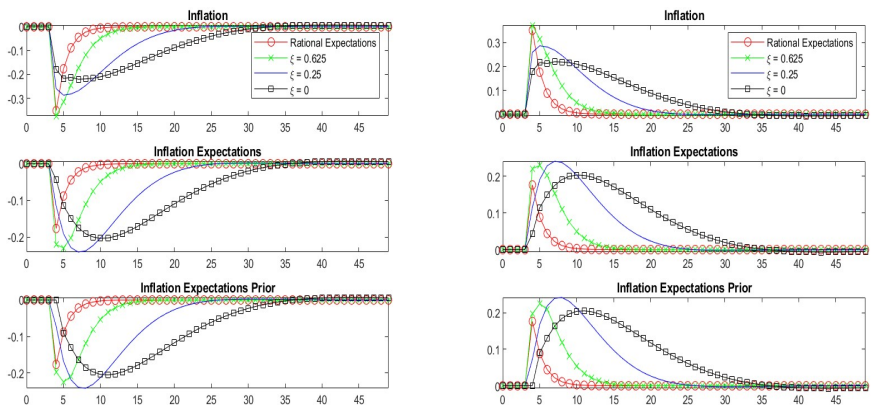
(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure A.2: Impulse Responses for inflation and its expectations from baseline model, $\xi = 0.25$ and $\eta_X = 0.5$, variation in credibility Γ



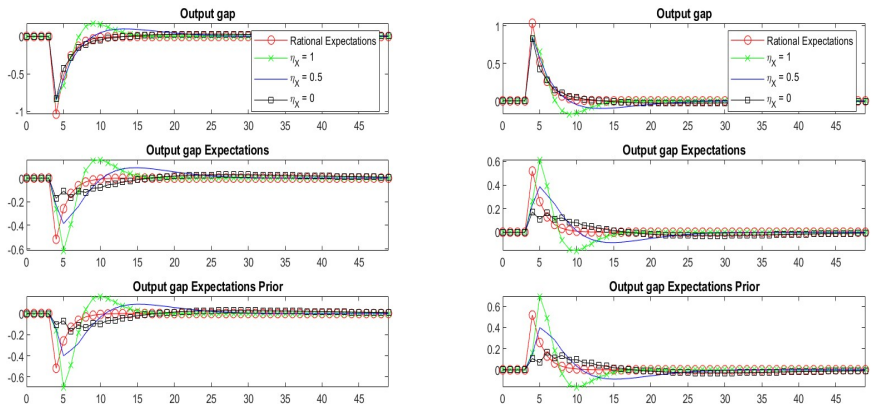
(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure A.3: Impulse Responses for output gap and its expectations from baseline model, credibility $\Gamma = 0.2$ and $\eta_X = 0.5$, variation in forward looking expectation component ξ



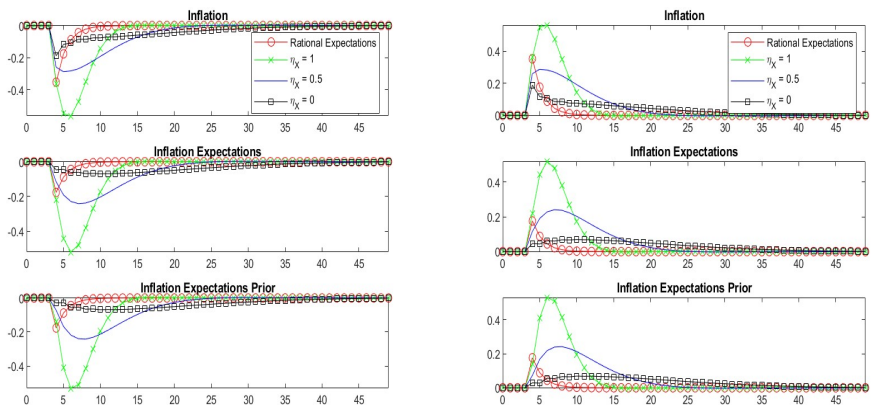
(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure A.4: Impulse Responses for inflation and its expectations from baseline model, credibility $\Gamma = 0.2$ and $\eta_X = 0.5$, variation in forward looking expectation component ξ



(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure A.5: Impulse Responses for output gap and its expectations from baseline model, credibility $\Gamma = 0.2$ and $\xi = 0.25$, variation in reaction to past forecast errors η_X



(a) 1 % Monetary Policy Shock (b) 1 % Demand Shock

Figure A.6: Impulse Responses for inflation and its expectations from baseline model, credibility $\Gamma = 0.2$ and $\xi = 0.25$, variation in reaction to past forecast errors η_X

B Appendix: Perfect Foresight Simulations at the ZLB

Set-up

In the perfect foresight setting with LMMCP, I compare the high cognitive discounting case ($\Gamma = 0.2, \xi = 0.25$) and the zero credibility case ($\Gamma = 0.0, \xi = 0.25$), varying η_X between 0, 0.5 and 1.

To quantify welfare loss, I use the second-order approximation to utility loss from Galí (2015), to denote per period loss as:

$$W_t = \frac{1}{2}(\frac{\kappa}{\lambda}y_t^2 + \frac{\epsilon}{\lambda}\pi_t^2) \tag{B.1}$$

In this set-up, I send the model to the ZLB using a 1 period negative demand shock of 10% and zero persistence: $\rho_g = 0$.

Results

With the 2nd specification from 3.2.2, I find that the IRF of inflation is the one that differs the most between the RE case and the 3 behavioural cases, both for $\Gamma = 0.2$ in B.1 and for $\Gamma = 0$ in $B.2$. This supports the findings from 4.2. From the graph for approximate Welfare loss per period in both figures, I again find that ZLB is the costliest in the $\eta_X = 1$ case, with the cost going down on lowering η_X. The curves for RE and $\eta_X = 0$ practically coincide, providing further evidence for the claims in 2.3.2.

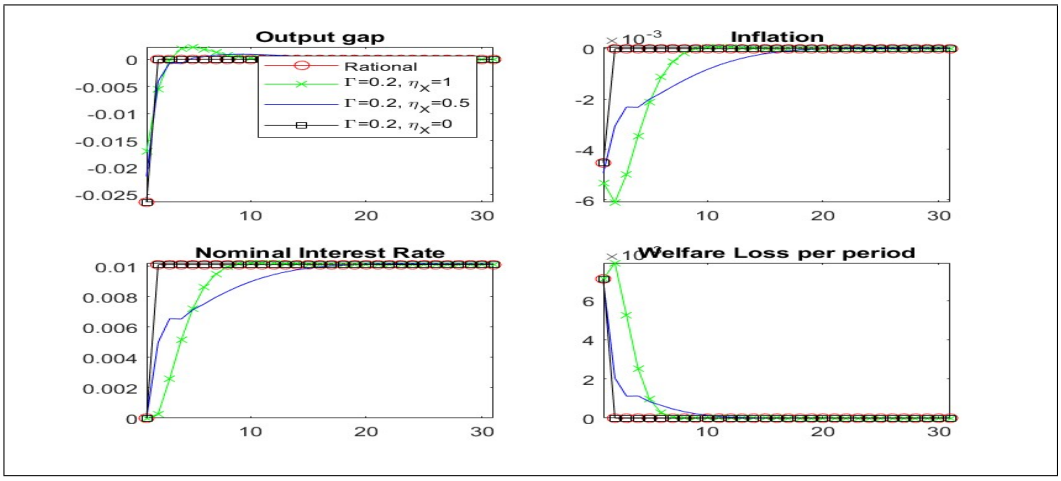


Figure B.1: Perfect Foresight Simulations of the Model sent to the ZLB with a demand shock at the high cognitive discounting benchmark ($\Gamma = 0.2, \xi = 0.25$), variation in expectations anchoring

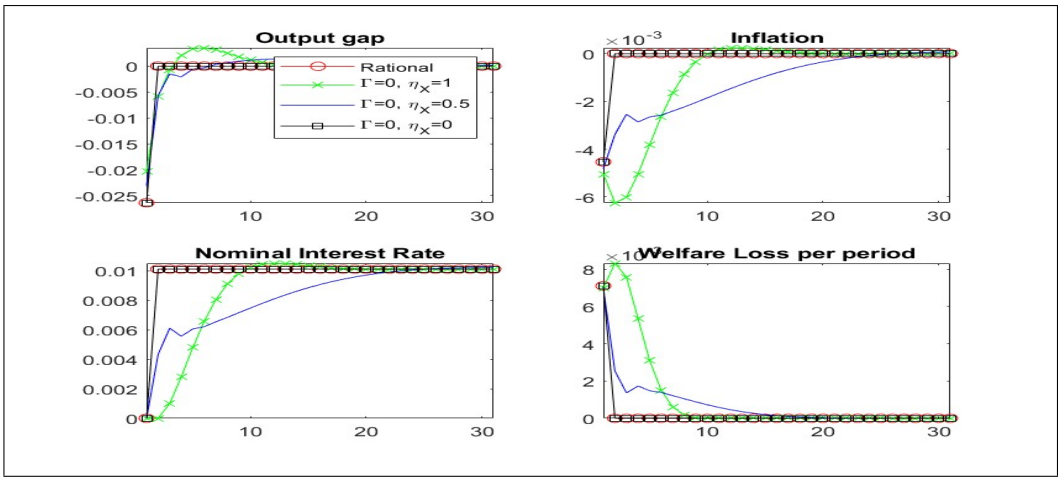


Figure B.2: Perfect Foresight Simulations of the Model sent to the ZLB with a demand shock at the zero credibility benchmark ($\Gamma = 0, \xi = 0.25$), variation in expectations anchoring

C Appendix: Robustness of Forward Guidance results from Perfect Foresight Simulations

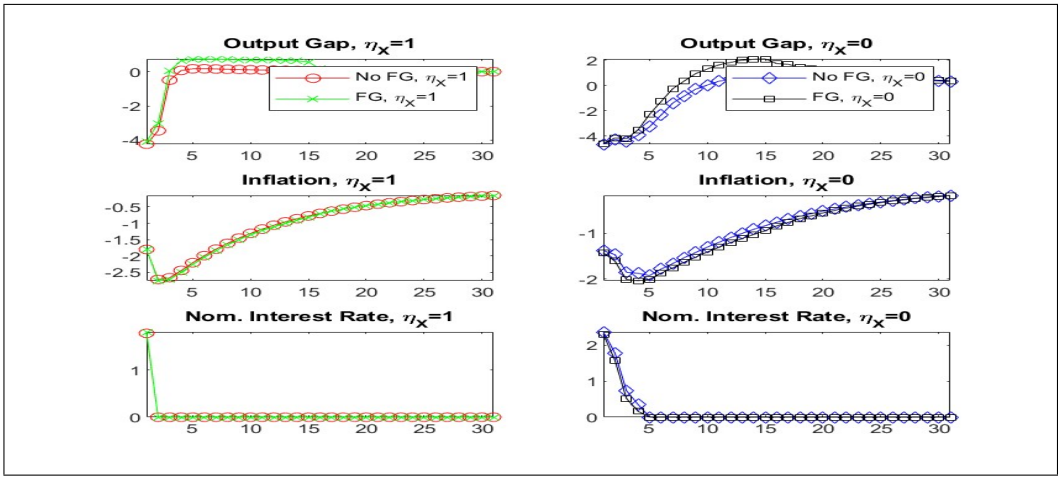


Figure C.1: Forward Guidance via CB expectations shock in response to MP shock, high cognitive discounting benchmark ($\Gamma = 0.2, \xi = 0.25$) comparing across η_X values

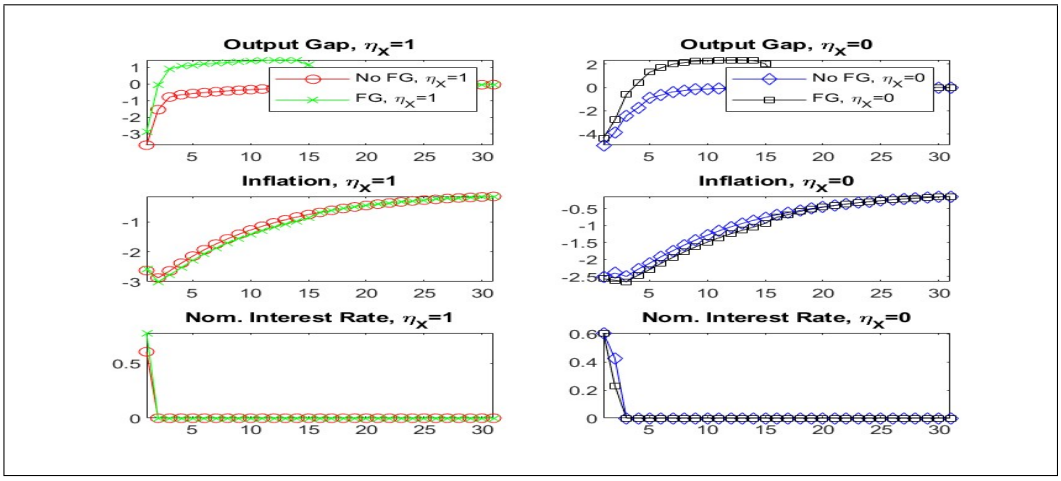


Figure C.2: Forward Guidance via CB expectations shock in response to MP shock, moderate cognitive discounting benchmark ($\Gamma = 0.6, \xi = 0.25$) comparing across η_X values